

Cambridge Checkpoint Mathematics

Grade 10 — Practice Examination

Mathematics

Grade 10 — Checkpoint-Style Practice Paper

Time allowed: 1 hour 30 minutes

Name: _____

Class: _____

Instructions

- Answer all questions.
- Show all necessary working.
- Give non-exact numerical answers to an appropriate degree of accuracy.
- You may use a calculator.
- The total number of marks is **60**.

Question	Marks
1	6
2	6
3	6
4	7
5	7
6	7
7	7
8	7
9	7
Total	60

Question 1 — Number

(a) Write 0.00056 in standard form.

[1]

(b) Calculate

$$\frac{3}{4} + \frac{5}{8}.$$

[2]

(c) A jacket originally costs \$240. Its price is reduced by 15%.
Calculate the sale price of the jacket.

[2]

(d) Round 7.38649 to three significant figures.

[1]

[6 marks]

Question 2 — Algebra

(a) Simplify

$$3x + 5 - 2x + 7.$$

[1]

(b) Expand and simplify

$$(x + 4)(x - 3).$$

[2]

(c) Factorise completely

$$6x^2 - 15x.$$

[2]

(d) Solve

$$4(2x - 1) = 3x + 11.$$

[1]

[6 marks]

Question 3 — Sequences and Functions

The first five terms of a sequence are

$$5, \quad 9, \quad 13, \quad 17, \quad 21.$$

(a) Write down the next two terms. [2]

(b) Find an expression for the n -th term of the sequence. [2]

(c) The function f is defined by

$$f(x) = 3x^2 - 2.$$

Find $f(4)$. [1]

(d) Solve

$$f(x) = 25.$$

[1]

[6 marks]

Question 4 — Ratio, Proportion and Percentages

A school has 840 students.

The ratio of students in Year 10 to students in Year 11 is

$$5 : 3.$$

(a) Calculate the number of Year 10 students. [2]

(b) 40% of the Year 10 students study French.
Calculate the number of Year 10 students who study French. [2]

(c) The number of Year 11 students increases by 12% the following year.
Calculate the new number of Year 11 students. [2]

(d) Explain why the answer in part (c) must be a whole number in this context. [1]

[7 marks]

Question 5 — Geometry

A triangle has side lengths 7 cm, 9 cm, and 12 cm.

- (a) Determine whether the triangle is right-angled.

Show all necessary working.

[2]

- (b) A different right-angled triangle has perpendicular sides 8 cm and 15 cm.

Calculate the length of the hypotenuse.

[2]

- (c) The angle between two sides of a triangle is 42° . The sides surrounding this angle have lengths 8 cm and 11 cm.

Calculate the area of the triangle.

[3]

[7 marks]

Question 6 — Coordinate Geometry

The points $A(-2, 5)$ and $B(4, -7)$ are shown on a coordinate grid.

- (a) Find the gradient of the line AB .

[2]

- (b) Find the equation of the line AB in the form

$$y = mx + c.$$

[2]

- (c) The line CD is perpendicular to AB .

Find the gradient of CD .

[1]

- (d) Line CD passes through the point $(2, 3)$.

Find its equation.

[2]

[7 marks]

Question 7 — Statistics

The following data show the number of minutes that 10 students spent completing a mathematical investigation.

18 21 15 24 19 27 21 16 23 26

- (a) Find the median. [2]
- (b) Find the range. [1]
- (c) Calculate the mean. [2]
- (d) A new student takes 60 minutes to complete the investigation.
Explain how adding this value affects the mean and the median. [2]

[7 marks]

Question 8 — Probability

A bag contains 5 red counters, 3 blue counters and 2 green counters.

A counter is selected at random.

(a) Find the probability that the counter is blue. [1]

(b) Find the probability that the counter is not green. [2]

Two counters are selected without replacement.

(c) Find the probability that both counters are red. [2]

(d) Find the probability that one counter is red and the other counter is blue. [2]

[7 marks]

Question 9 — Problem Solving

A company manufactures rectangular storage boxes.

Each box has length x cm, width 8 cm and height 5 cm.

(a) Write an expression for the volume V of the box. [1]

(b) The volume of one box is 360 cm^3 .
Find the value of x . [2]

(c) The company wants to increase the length by 25% while keeping the width and height unchanged.
Find the new volume of the box. [2]

(d) The boxes are packed into a larger rectangular container. The container has dimensions 60 cm, 40 cm and 30 cm.
Calculate the volume of the container. [1]

(e) The company claims that 40 of the boxes can fit inside the container.
Use the volumes to determine whether this claim is possible. Explain your answer. [1]

[7 marks]

End of Paper